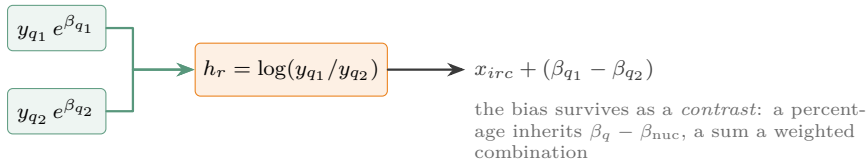
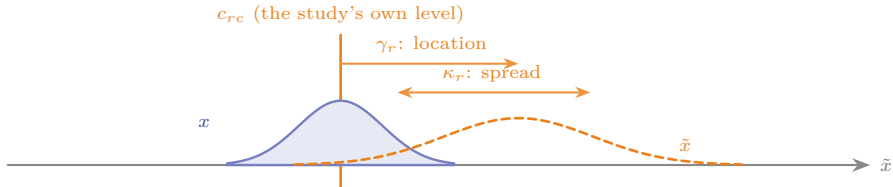


upstream of h_r : a bias on the *species*



downstream of h_r : an affine map on the *row*, pivoting at c_{rc}



$\gamma_r = Z_r^\top a$ and $\log \kappa_r = Z_r^\top b$ read attributes of the readout with no mechanistic counterpart (assay modality; density, fraction or concentration). They act on the finished row, so they do not decompose through a ratio.